EWU NATIONAL ROBOFEST 2026

DOMINATION IS DESTINY

ORGANIZED BY

EAST WEST UNIVERSITY ROBOTICS CLUB

Programming Hero

Presents

NATIONAL Hackathon

EVENT RULEBOOK

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Introduction

Welcome to the NRF26 Hackathon, a flagship segment of National Robo Fest 2026, designed to foster innovation, creativity, and problem-solving through technology.

The competition will be conducted in two stages:

- **Stage 1:** Idea Submission & Initial Evaluation: Teams submit project proposals, from which selected teams advance to the final round.
- **Stage 2:**Hackathon Phase & Final Presentation: Shortlisted teams will build and present their solutions within a limited time, evaluated by a panel of expert judges.

The NRF26 Hackathon encourages collaboration, technical excellence, and real-world innovation among young technologists.

Eligibility & Team Requirements

Eligible Participants

Participation is open to:

- University students
- Polytechnic institute students
- College-level students or equivalent institutions

All participants must possess a **valid student ID card.**

Team Formation

- Each team must consist of **2–3 members.**
- One member must be assigned as the **Team Leader.**
- A participant may join **only one team.**
- Cross-institution teams are allowed.
- All official communication will be conducted through the Team Leader.

General Rules

- Participants must maintain professional conduct throughout the event.
- Any form of plagiarism, cheating, or unethical behavior will result in immediate disqualification.
- Submitted projects must be original work developed by the team.
- Public APIs, open-source libraries, and datasets are permitted.
- Organizers reserve the right to verify project authenticity at any stage.
- Judges' decisions will be **final and binding.**
- The organizing committee reserves the right to modify rules, schedule, or format if necessary.

Problem Alignment

The **NRF26 Hackathon** will follow a **Mission-Based Challenge** format. Instead of developing a project based on a single idea, participating teams will be required to design a solution that addresses **multiple real-world challenges simultaneously.**

The hackathon challenge will be divided into **three core missions**, each representing a major problem domain inspired by global development priorities. Teams must develop **one integrated solution** that incorporates elements addressing **all three missions**.
This structure encourages participants to think beyond isolated solutions and build **comprehensive technological systems** capable of solving interconnected real-world problems.

Mission 1: EarthCare

Focus: Climate, environment, and sustainability.

Teams must include a component in their solution that contributes to environmental protection or climate-related problem solving.

Example solution areas may include:

- Flood prediction or disaster alert systems
- Smart waste management platforms
- Air or water quality monitoring solutions
- Energy usage monitoring and sustainability dashboards

The objective of this mission is to explore how technology can help **protect the environment and promote sustainable development.**

Mission 2: HealthNet

Focus: Healthcare accessibility and wellbeing.

Teams must incorporate features that improve healthcare access, awareness, or support for communities.

Possible solution directions include:

- AI-powered symptom checking or medical assistance tools
- Platforms connecting patients with nearby hospitals or clinics
- Mental health support systems
- Medicine availability tracking platforms

The aim of this mission is to develop technological solutions that **enhance healthcare accessibility and wellbeing.**

Mission 3: SmartCity

Focus: Urban innovation and smart infrastructure.

Participants must develop a component that addresses challenges faced in modern cities and improves urban life using technology.

Example solution areas include:

- Smart traffic monitoring and route optimization systems
- Public issue reporting platforms for city management
- Emergency alert and safety systems
- Smart transportation and navigation solutions

This mission focuses on building technologies that contribute to **efficient, safe, and intelligent urban environments.**

Mission Completion Requirement

Each participating team must develop **one integrated project** that addresses **all three missions:**

EarthCare, HealthNet, and SmartCity.

Teams are expected to design solutions where different components of the system contribute to fulfilling these missions simultaneously.

For example, a project may combine:

- **EarthCare:** Environmental monitoring or disaster prediction
- **HealthNet:** Emergency medical assistance or healthcare access
- **SmartCity:** City-wide alert systems or urban management tools

Projects that successfully demonstrate meaningful integration of **all three missions** will be considered more competitive during evaluation.

Application Phase

- Participants must form teams before applying.
- Teams must complete registration through the **official online application form**.
- Each team must submit:
  - Project/Application Name
  - Idea Description
  - Proposed Technology Stack(optional but recommended)

- The **online application round is free of cost**.
- Teams will undergo preliminary screening based on innovation, feasibility, and relevance.
- Selected teams will receive confirmation via email.

Offline Confirmation

- Selected teams must confirm participation by completing registration payment within the specified deadline.
- Registration Fee: **BDT 500 per participant**.
- Failure to complete payment within deadline will result in slot cancellation and reassignment to waiting-list teams

Hackathon Phase

Reporting & Onboarding

- Selected teams must report at the venue on **[Event Date]** within the official reporting time.
- Participants must bring valid student ID cards.
- During onboarding, participants will receive:
  - Event T-shirt
  - Participant ID Card
  - Goodies
  - Lunch & Snacks
  - Assigned Workspace

Development Phase

- Teams will develop a **Web Application** or **Web + AI–based Solution** based on their submitted idea.
- The development duration will be **10 hours**.
- A **30-minute lunch break** will be provided outside hacking time.
- Teams must bring their own devices (laptop, mobile, chargers, extensions, etc.).
- Internet access will be provided; backup hotspot is recommended.
- Each team must bring their own required devices and accessories necessary for development, including **laptops, chargers, mobile devices, extension boards/multiplugs, adapters, USB hubs (for Mac users), and any additional peripherals** required for their project.

Technical Rules

- No restriction on programming languages or frameworks.
- Low-code / No-code platforms are discouraged unless significant custom development is demonstrated.
- Final output must be:
  - Functional
  - Viewable in a web browser
  - Demonstrable during judging

Bonus consideration will be given for:

- AI Integration
- Innovative Features
- Real-world Applicability

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Evolution of Dreams

Judging & Evaluation Criteria

Criteria	Description	Marks
Problem Relevance	Alignment with SDGs & real-world need	10
Innovation & Creativity	Originality of concept	15
Technical Implementation	Complexity & engineering quality	20
Functionality	Working features & stability	15
UI/UX & Usability	User experience & design	10
Impact & Scalability	Practical usefulness & future scope	10
Team Collaboration	Workflow & time management	10
Presentation & Demo	Communication & clarity	10
Total		**100**

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Registration Fees

Phase	Fee
Online Application	Free
Offline Hackathon	500tk per participant

Presentation & Final Judging

- Each team will receive:
  - **3 minutes** for presentation/demo
  - **2 minutes** for Q&A session
- Judges may inspect source code if required.
- Top teams will be selected based on total evaluation scores.

Prize Pool Format (Sample — editable)

Total Prize Pool: [To Be Announced]

Champion - TBA

1st Runner-Up - TBA

2nd Runner-Up - TBA

Special Awards:

- Best Innovation Award
- Best AI Solution
- Judges' Choice Award

Certificates will be awarded to all on-site participants.

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Code of Conduct & Organizer Rights

- Participants must maintain discipline, respect, and ethical behavior.
- Harassment, misconduct, or rule violation will lead to disqualification.
- Organizers reserve the right to:
  - Modify schedules
  - Update rules
  - Postpone or cancel the event under unforeseen circumstances.

Important Note

- The organizing committee will not provide accommodation or overnight staying facilities for participants.
- Participants are requested to arrange their own accommodation and transportation if required.